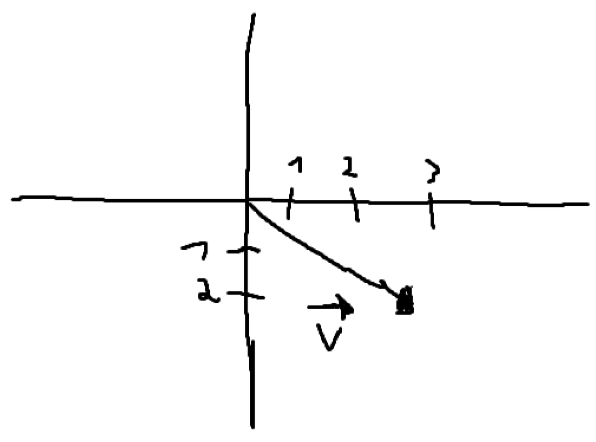
Tanjic 2022189

1. Representação no plano

Represente graficamente no plano o vetor

$$\vec{v} = (3, -2).$$



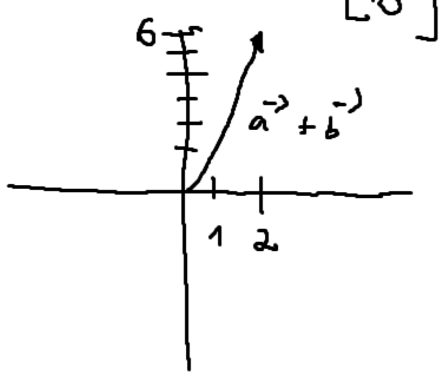
2. Soma de vetores em $\mathbb{R}^2$

Calcule

$$(4, 1) + (-2, 5).$$

$\vec{a} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$

$\vec{a} + \vec{b} = \begin{bmatrix} 4-2 \\ 1+5 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$



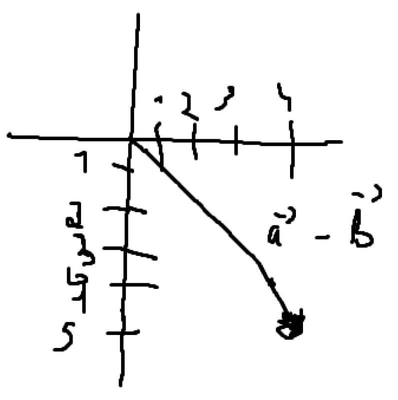
3. Subtração de vetores em $\mathbb{R}^2$

Calcule

$$(7, -1) - (3, 4).$$

$\vec{a} = \begin{bmatrix} 7 \\ -1 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$\vec{a} - \vec{b} = \begin{bmatrix} 7-3 \\ -1-4 \end{bmatrix} = \begin{bmatrix} 4 \\ -5 \end{bmatrix}$



4. Multiplicação por escalar

Calcule

$$3(2, -3) \quad \text{e} \quad -2(1, 4).$$

$\vec{a} = \begin{bmatrix} 2 \\ -3 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$3 \times \vec{a} = 3 \times \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \times 2 \\ 3 \times -3 \end{bmatrix} = \begin{bmatrix} 6 \\ -9 \end{bmatrix}$

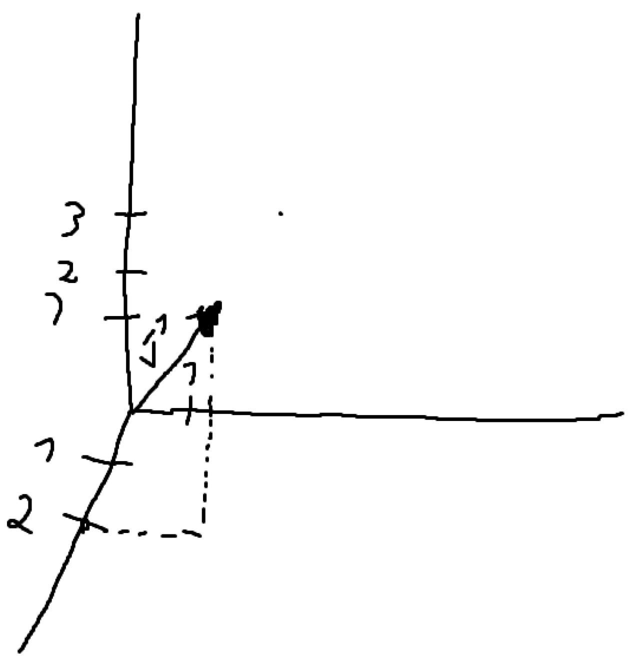
$-2 \times \vec{b} = -2 \times \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \times 1 \\ -2 \times 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \end{bmatrix}$

5. Representação em $\mathbb{R}^3$

Represente o vetor

$$\vec{u} = (2, 1, 3)$$

num sistema tridimensional, indicando sua projeção nos eixos.



6. Soma em $\mathbb{R}^3$

Calcule

$$(1, 4, -2) + (3, -1, 5).$$

$\vec{a} + \vec{b} = \begin{bmatrix} 1+3 \\ 4-1 \\ -2+5 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 3 \end{bmatrix}$

$\vec{a} = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$

1.

Dado $u = (3, -1)$ e $v = (2, 4)$, calcule:

a) $u + v$

b) $u - v$

c) $2u$

a) $\vec{u} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$u + v = \begin{bmatrix} 3+2 \\ -1+4 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ //

b) $\begin{bmatrix} 3-2 \\ -1-4 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$ //

c) $2 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \times 3 \\ 2 \times (-1) \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$ //

2.

Calcule a norma do vetor $v = (4, -3)$:

$$\|v\| = ?$$

$\|v\| = \sqrt{x^2 + y^2}$

$\|v\| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$

3.

Dado o vetor $w = (1, 2, -2)$, calcule:

a) $3w$

b) $-w$

c) $\|w\|$

a) $\vec{w} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$

$3\vec{w} = 3 \times \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{matrix} 3 \times 1 \\ 3 \times 2 \\ 3 \times (-2) \end{matrix} = \begin{bmatrix} 3 \\ 6 \\ -6 \end{bmatrix}$

b) $-w = -1 \times \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$

$-w = \begin{bmatrix} -1 \times 1 \\ -1 \times 2 \\ -1 \times (-2) \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}$

c) $\|w\| = \sqrt{x^2 + y^2 + z^2}$

$\|w\| = \sqrt{1^2 + 2^2 + (-2)^2}$

$= \sqrt{1 + 4 + 4}$

$= \sqrt{9}$

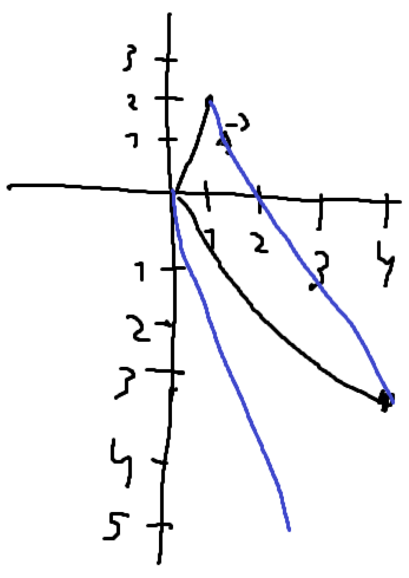
$= 3$

4.

Determine o vetor que vai de $A = (1, 2)$ até $B = (4, -3)$.

$\vec{B} - \vec{A} = \begin{bmatrix} 4-1 \\ 2+3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$\vec{A} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{B} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$



5.

Dados os vetores $u = (1, 0, 2)$ e $v = (-2, 3, 1)$, calcule:

a) $u + v$

b) $u - 2v$

$u = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \quad v = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$

a) $u + v = \begin{bmatrix} 1-2 \\ 0+3 \\ 2+1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 3 \end{bmatrix}$

b) $u - 2v = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \times (-2) \\ 2 \times 3 \\ 2 \times 1 \end{bmatrix}$

$= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} -4 \\ 6 \\ 2 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 0-6 \\ 2-2 \end{bmatrix}$

$= \begin{bmatrix} 5 \\ -6 \\ 0 \end{bmatrix}$

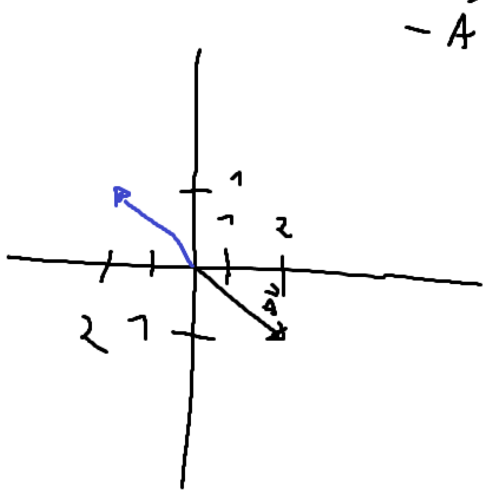
6.

No plano, represente geometricamente o vetor $(2, -1)$.

Em seguida, determine seu vetor oposto.

$\vec{A} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$-\vec{A} = -1 \times \begin{bmatrix} 2 \\ -1 \end{bmatrix}$



7.

Se $u = (a, 3)$ e $2u = (6, 6)$, determine o valor de a e o vetor u.

$u = \begin{bmatrix} a \\ 3 \end{bmatrix} \quad 2u = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$

$2u = \begin{bmatrix} 2 \times a \\ 2 \times 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$

$a = 3$

$\frac{\cancel{2}a}{\cancel{2}} = \frac{6}{2} = \boxed{3}$